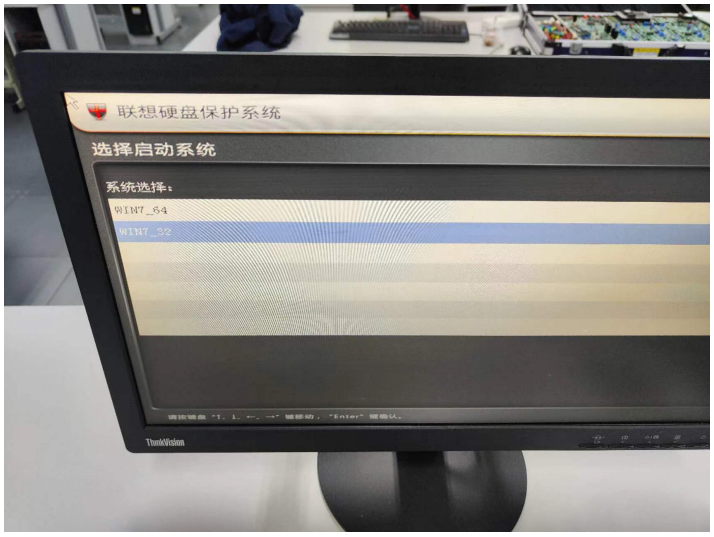
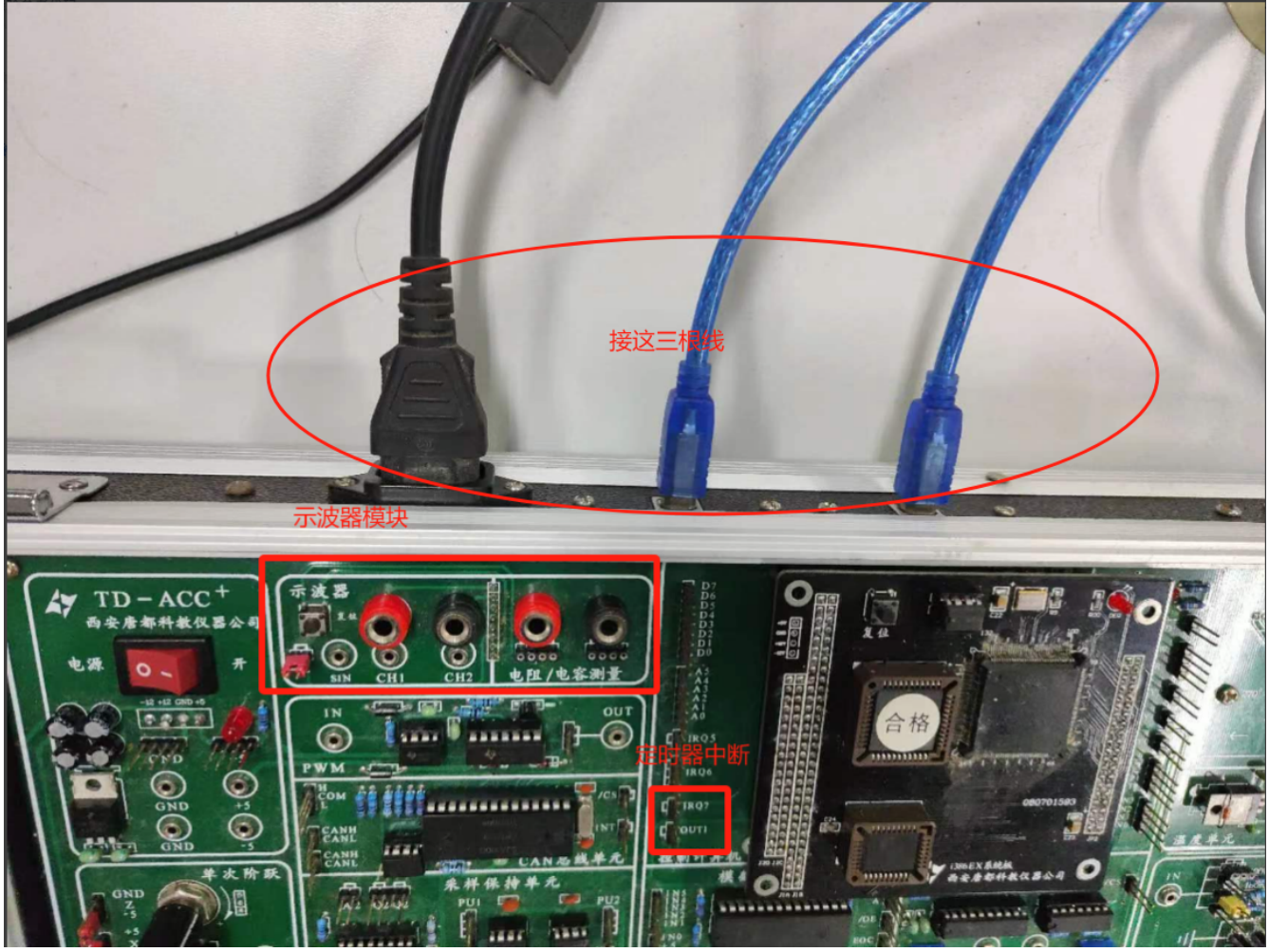
计算机控制技术实验参考代码

指导书 https://sy666.lanzoue.com/iDmde1xru30f

系统选择 32 位



实验箱接线



实验一 系统认识及程序调试练习

实验程序 1（采用软件延时方式）

Plain Text

```
 1 CODE    SEGMENT
 2         ASSUME CS:CODE
 3 START:  MOV CX,001AH
 4         MOV AH,01
 5         MOV AL,13
 6         INT 10H
 7         CALL DELAY
 8         MOV AL,41H
 9 AGAIN:  MOV AH,01
10         INT 10H
11         INC AL
12         LOOP AGAIN
13         JMP START
14 DELAY:  PUSH CX
15         MOV CX,0FFFFH
16 DEL1:   PUSH AX
17         POP AX
18         LOOP DEL1
19         POP CX
20         RET
21 CODE    ENDS
22         END START
```

实验程序2（采用定时中断方式）

先连线！！！！ 定时器中断模块

BX的值修改在第20、29行

Plain Text

```
CODE    SEGMENT
        ASSUME CS:CODE
START:  MOV AX,OFFSET IRQ7
        MOV SI,003CH
        MOV [SI],AX
        MOV AX,CS
        MOV SI,003EH
        MOV [SI],AX
        CLI
        CALL SYSINTI
        MOV DX,0F043H
        MOV AL,076H
        OUT DX,AL
        MOV DX,0F041H
        MOV AL,10H
        OUT DX,AL
        MOV DX,0F041H
        MOV AL,27H
        OUT DX,AL
        MOV BX,64H
        MOV AH,01
        MOV AL,13
        INT 10H
AGAIN:  STI
        HLT
        JMP AGAIN
IRQ7:   DEC BX
        JNZ FINISH
        MOV BX,64H
        MOV CX,001AH
        MOV AL,41H
AGAIN1: MOV AH,01
        INT 10H
        INC AL
        LOOP AGAIN1
        MOV AH,01
        MOV AL,13
        INT 10H
FINISH: MOV AL,20H
        OUT 20H,AL
        IRET

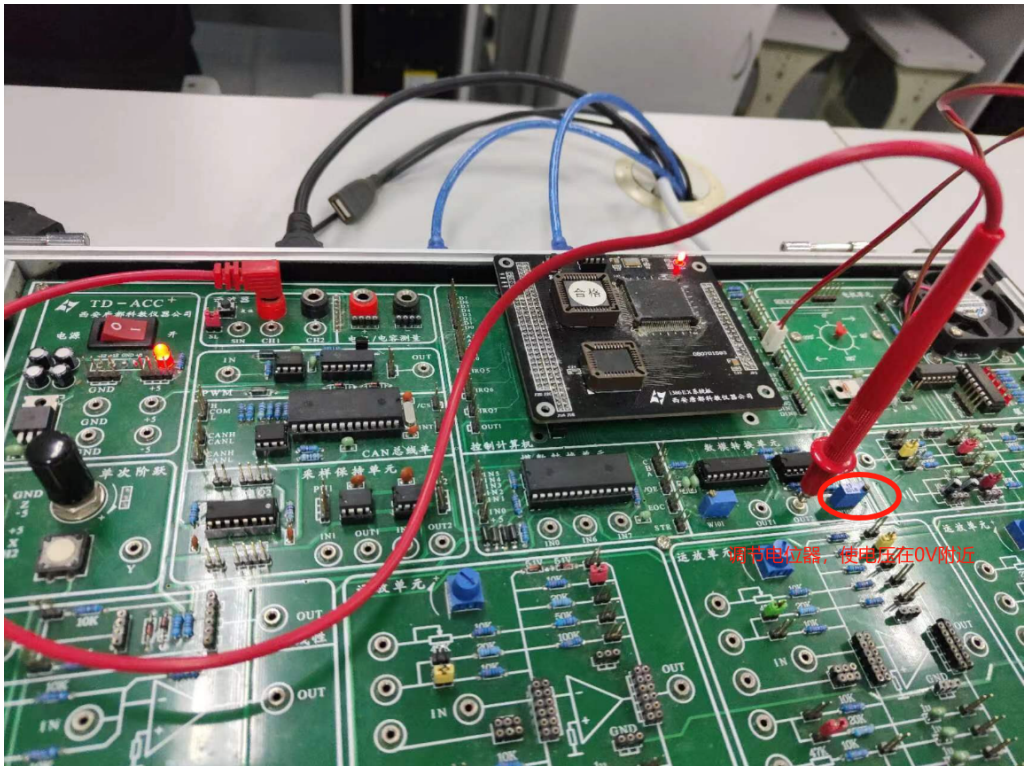
SYSINTI:MOV   AX,8000H          ;系统初始化子程序
```

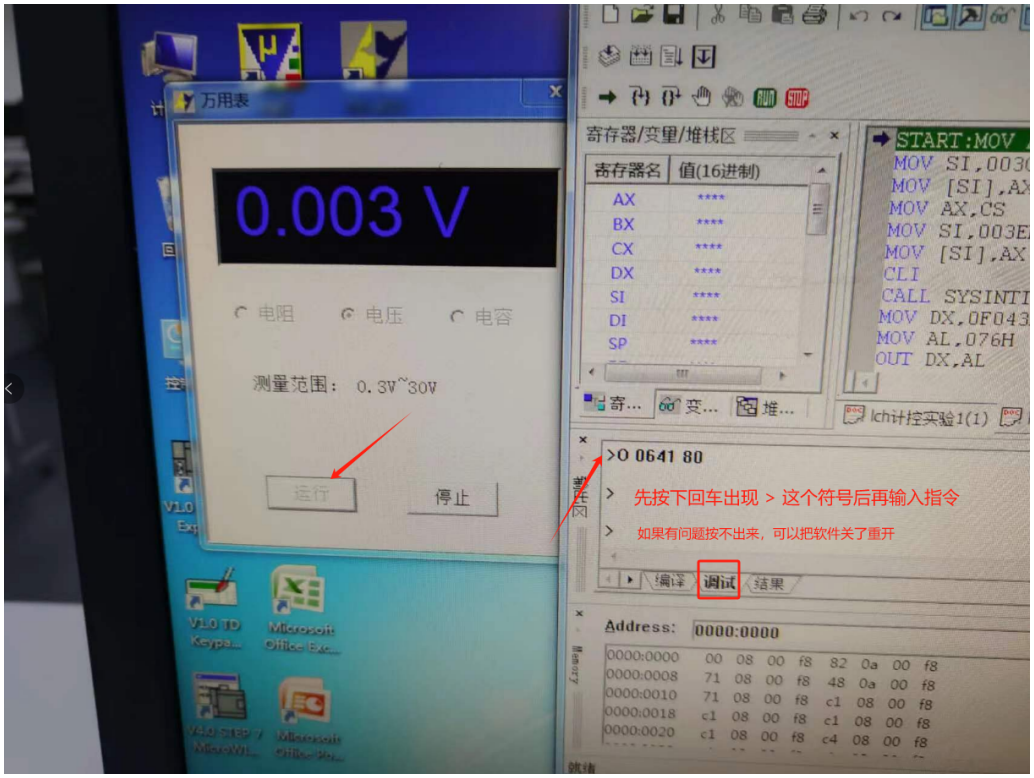
```
44          OUT    23H,AL              ;扩展IO使能
45          XCHG   AL,AH
46          OUT    22H,AL
47          OUT    22H,AX
48
49          MOV     DX,0F822H          ;初始化管脚配置P2CFG，配置CS0#
50          MOV     AL,70H
51          OUT     DX,AL
52          MOV     DX,0F824H          ;初始化管脚配置P3CFG,配置主片IRQ7
53          MOV     AL,0B2H
54          OUT     DX,AL
55          MOV     DX,0F832H          ;初始化管脚配置INTCFG
56          MOV     AL,0AH
57          OUT     DX,AL
58          MOV     DX,0F834H          ;初始化管脚配置TMRCFG
59          MOV     AL,15H              ;将GATE1接VCC
60          OUT     DX,AL
61          MOV     AL,11H              ;初始化主片8259
62          OUT     20H,AL
63          MOV     AL,08H
64          OUT     21H,AL
65          MOV     AL,04H
66          OUT     21H,AL
67          MOV     AL,01H
68          OUT     21H,AL
69          MOV     AL,6FH              ;写主片8259的中断屏蔽字，允许主片的IRQ7(用户
   程序使用)和IRQ4(系统通讯用)
70          OUT     21H,AL
71          MOV     AL,11H              ;初始化从片8259
72          OUT     0A0H,AL
73          MOV     AL,30H
74          OUT     0A1H,AL
75          MOV     AL,02H
76          OUT     0A1H,AL
77          MOV     AL,01H
78          OUT     0A1H,AL
79          MOV     AL,0FFH
80          OUT     0A1H,AL
81          RET
82 CODE     ENDS
83          END     START
```

实验二 D/A 转换实验

先调电位

以0641（OUT2）为例，0640（OUT1）同理





实验程序1（采用软件延时方式）

Plain Text

```
CODE      SEGMENT
          ASSUME    CS:CODE
START:    MOV    AL,00H
          MOV    BL,00H
AGAIN:    MOV    DX,0640H
          OUT    DX,AL
          XCHG   AL,BL
          MOV    DX,0641H
          OUT    DX,AL
          XCHG   AL,BL
          CALL   DELAY
          INC    AL          ;变量1加1 锯齿波
          JNZ    AGAIN       ;变量1不等于0跳转
          NOT    BL            ;变量2取反
          JMP    AGAIN
DELAY:    PUSH   CX            ;延时子程序
          MOV    CX,00FFFH
DEL1:     PUSH   AX
          POP    AX
          LOOP   DEL1
          POP    CX
          RET
CODE      ENDS
          END    START
```

实验程序 2（采用定时中断方式）

先连线！！！！ 定时器中断模块

定时时间 1ms，定时器常数 1000 转成 16 进制 03e8H，所以定时器常数高字节是 03H，低字节 0E8H，这个与实验一中的定时器常数不一样，注意区别

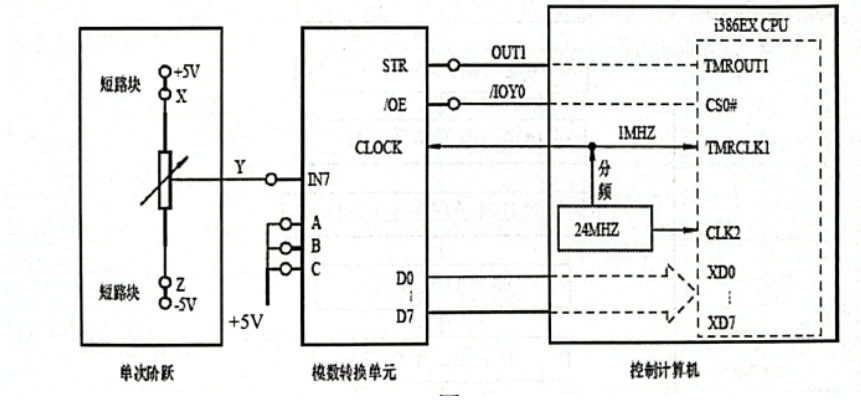
Plain Text

```
CODE SEGMENT
      ASSUME CS:CODE
START :MOV        AX, OFFSET IRQ7;填写8259的7号中断矢量入口地址的偏移量
       MOV        SI,003CH        ;填写8259中断7的中段矢量
       MOV        [SI],AX           ;填偏移量矢量
       MOV        AX,CS            ;填写8259中断矢量入口地址的段地址
       MOV        SI,003EH
       MOV        [SI],AX
       CLI                         ;关系统总中断
       CALL       SYSINTI           ;调用系统初始化子程序
       MOV        DX,0F043H  ;定时器的TMRCON的地址
       MOV        AL,076H           ;初始化1＃定时器定时1ms
       OUT        DX, AL
       MOV        DX,0F041H        ;写1#定时器定时常数的低字节
       MOV        AL,0E8H
       OUT        DX,AL
       MOV        DX,0F041H        ;写1#定时器定时常数的高字节
       MOV        AL,03H
       OUT        DX,AL
       MOV        CL,00H            ;变量1、2=0
       MOV        BL,00H
AGAIN:STI           ;打开系统总中断
       HLT                      ;停机等待直到有中断产生
       JMP        AGAIN            ;中断返回后继续
IRQ7:  MOV        AL,CL
       MOV        DX,0640H
       OUT        DX,AL
       MOV        AL,BL
       MOV        DX,0641H
       OUT        DX,AL
       INC        CL;变量1加一
       JNZ        FINISH ;变量1不等于0跳转
       NOT        BL;变量2取反
FINISH:MOV        AL,20H             ;写中断结束字
       OUT        20H,AL
       IRET                          ;中断返回
SYSINTI: MOV   AX,8000H;系统初始化子程序
        OUT        23H,AL    ;扩展IO使能
        XCHG        AL,AH
        OUT        22H,AL
        OUT        22H,AX
        MOV        DX,0F822H    ;初始化管脚配置P2CFG，配置CS0#
        MOV        AL,70H
```

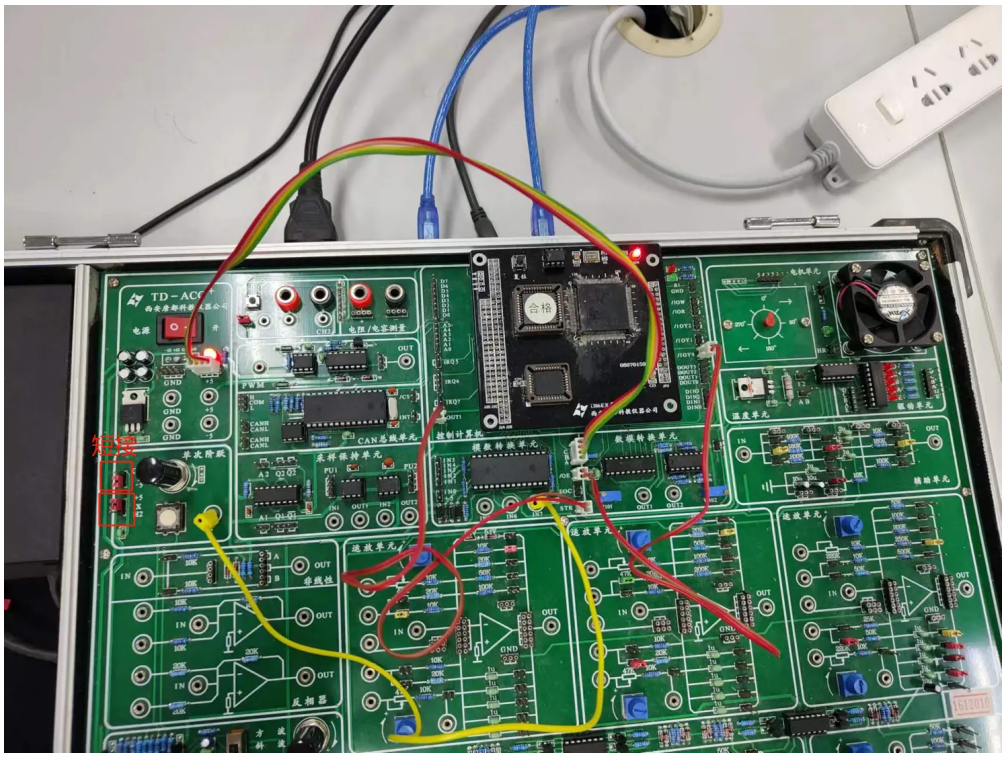
```
44          OUT         DX,AL
45          MOV         DX,0F824H           ;初始化管脚配置P3CFG,配置主片IRQ7
46          MOV         AL,0B2H
47          OUT         DX,AL
48          MOV         DX,0F832H           ;初始化管脚配置INTCFG
49          MOV         AL,0AH
50          OUT         DX,AL
51          MOV         DX,0F834H           ;初始化管脚配置TMRCFG
52          MOV         AL,15H               ;将GATE1接VCC
53          OUT         DX,AL
54          MOV         AL,11H               ;初始化主片8259
55          OUT         20H,AL
56          MOV         AL,08H
57          OUT         21H,AL
58          MOV         AL,04H
59          OUT         21H,AL
60          MOV         AL,01H
61          OUT         21H,AL
62          MOV         AL,6FH               ;写主片8259的中断屏蔽字，允许主片的
   I
63          OUT         21H,AL
64          MOV         AL,11H               ;初始化从片8259
65          OUT         0A0H,AL
66          MOV         AL,30H
67          OUT         0A1H,AL
68          MOV         AL,02H
69          OUT         0A1H,AL
70          MOV         AL,01H
71          OUT         0A1H,AL
72          MOV         AL,0FFH
73          OUT         0A1H,AL
74          RET
75
76 CODE     ENDS
77           END         START
```

实验三 A/D 转换实验

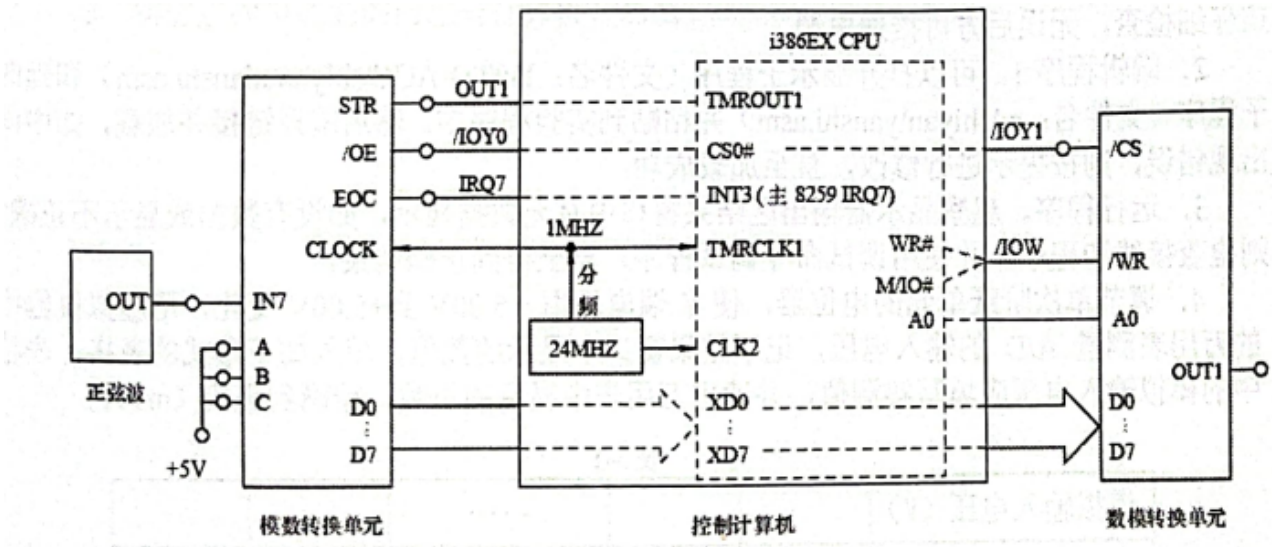
第一个接线



实物接线参考



第二个接线



实物图略

重新接线，注意区分！！！

内容一：

1ms,03H,E8H

Plain Text

```
CODE SEGMENT
ASSUME CS:CODE
START: CALL SYSINTI
       MOV DX,0F043H
       MOV AL,076H
       OUT DX,AL
TIME:  MOV DX,0F041H
       MOV AL,0E8H
       OUT DX,AL
       MOV DX,0F041H
       MOV AL,03H
       OUT DX,AL
       CALL DELAY1
       MOV DX,0600H
       IN AL,DX
       CALL DSPY
       CALL DELAY2
       JMP TIME
SYSINTI:MOV AX,8000H
       OUT 23H,AL
       XCHG AL,AH
       OUT 22H,AL
       OUT 22H,AX
       MOV DX,0F822H
       MOV AL,70H
       OUT DX,AL
       MOV DX,0F824H
       MOV AL,0B2H
       OUT DX,AL
       MOV DX,0F832H
       MOV AL,0AH
       OUT DX,AL
       MOV DX,0F834H
       MOV AL,15H
       OUT DX,AL
       MOV AL,11H
       OUT 20H,AL
       MOV AL,08H
       OUT 21H,AL
       MOV AL,04H
       OUT 21H,AL
       MOV AL,01H
       OUT 21H,AL
```

```
44        MOV AL,6FH
45        OUT 21H,AL
46        MOV AL,11H
47        OUT 0A0H,AL
48        MOV AL,30H
49        OUT 0A1H,AL
50        MOV AL,02H
51        OUT 0A1H,AL
52        MOV AL,01H
53        OUT 0A1H,AL
54        MOV AL,0FFH
55        OUT 0A1H,AL
56        RET
57 DSPY:     PUSH  AX
58           MOV   AX,0001
59           INT   10H
60             POP   AX
61           MOV   CL,04
62           MOV   DL,AL
63           SHR   AL,CL
64           CMP   AL,09
65           JNA   DLAY1
66           ADD   AL,07
67           NOP
68 DLAY1:    ADD   AL,30H
69           MOV   AH,01
70           INT   10H
71           AND   DL,0FH
72           CMP   DL,09
73           JNA   DLAY2
74           ADD   DL,07
75 DLAY2:    ADD   DL,30H
76           MOV   AL,DL
77           MOV   AH,01
78           INT   10H
79           RET
80 DELAY1:PUSH CX
81        MOV CX,0FFFFH
82 DEL1:  PUSH AX
83        POP AX
84        LOOP DEL1
85        POP CX
86        RET
87 DELAY2:PUSH CX
88        PUSH BX
89        MOV BX,300
```

```
 90 CZ:    MOV CX,0FFFFH
 91 DEL2:  PUSH AX
 92        POP AX
 93        LOOP DEL2
 94        DEC BX
 95        JZ CZ
 96        POP CX
 97        POP BX
 98        RET
 99 CODE ENDS
100 END START
```

内容二：

10ms,2710H

修改第七行Tk采样周期的值！

Plain Text

```
AD_ADDR      EQU         0600H
DA1_ADDR  EQU        0640H
STACK1    SEGMENT       STACK
          DW            256 DUP(?)
STACK1    ENDS
DATA      SEGMENT
TK        DB      04H        ;采样周期
DATA      ENDS
CODE      SEGMENT
          ASSUME    CS:CODE, DS:DATA
START:    MOV     AX,DATA
          MOV     DS,AX
          PUSH    DS
          MOV     AX,0000H
          MOV     DS,AX

          MOV     AX,OFFSET IRQ7;填写8259的7号中断矢量入口地址的偏移量
          MOV     SI,003CH        ;填写8259中断7的中段矢量
          MOV     [SI],AX        ;填偏移量矢量
          MOV     AX,CS            ;填写8259中断矢量入口地址的段地址
          MOV     SI,003EH        ;填写7号中断段地址矢量
          MOV     [SI],AX
          POP     DS
          CLI                     ;关系统总中断
          CALL    SYSINTI         ;调用系统初始化子程序
          MOV     DX,0F043H
          MOV     AL,074H       ;初始化1#定时器定时10ms

          OUT     DX,AL
          MOV     DX,0F041H
          MOV     AL,10H           ;写1#定时器定时常数的低字节

          OUT     DX,AL
          MOV     DX,0F041H
          MOV     AL,27H           ;写1#定时器定时常数的高字节,同时启动AD转
   换
          OUT     DX,AL
          MOV     BL,TK         ;赋采样周期变量初值
AGAIN:    STI                     ;打开系统总中断
          HLT                     ;停机等待直到有中断产生

          JMP     AGAIN            ;继续

```

```
IRQ7:     DEC     BL              ;采样周期到否
          JNZ     FINISH          ;采样周期没到，中断返回
          MOV      DX,AD_ADDR
          IN      AL,DX           ;读AD采样值
          MOV      DX,DA1_ADDR
          OUT      DX,AL
          MOV     BL,TK
FINISH:   MOV     AL,20H          ;中断结束
          OUT     20H,AL
          IRET                      ;中断返回
SYSINTI:  MOV   AX,8000H          ;系统初始化子程序
          OUT   23h,AL              ;扩展IO使能
          XCHG  AL,AH
          OUT   22H,AL
          OUT   22H,AX

          MOV    DX,0F822H          ;初始化管脚配置P2CFG，配置CS0#
          MOV    AL,70H
          OUT    DX,AL
          MOV    DX,0F824H          ;初始化管脚配置P3CFG,配置主片IRQ7
          MOV    AL,0B2H
          OUT    DX,AL
          MOV    DX,0F832H          ;初始化管脚配置INTCFG
          MOV    AL,0AH
          OUT    DX,AL
          MOV    DX,0F834H          ;初始化管脚配置TMRCFG
          MOV    AL,15H              ;将GATE1接VCC
          OUT    DX,AL

          MOV    AL,11H              ;初始化主片8259
          OUT    20H,AL
          MOV    AL,08H
          OUT    21H,AL
          MOV    AL,04H
          OUT    21H,AL
          MOV    AL,01H
          OUT    21H,AL
          MOV    AL,6FH              ;写主片8259的中断屏蔽字，允许主片的IRQ7
(用户程序使用)和IRQ4(系统通讯用)
          OUT    21H,AL

          MOV    AL,11H              ;初始化从片8259
          OUT    0A0H,AL
          MOV    AL,30H
          OUT    0A1H,AL
          MOV    AL,02H
```

```
85              OUT     0A1H,AL
86              MOV     AL,01H
87              OUT     0A1H,AL
88              MOV     AL,0FFH
89              OUT     0A1H,AL
90              RET
91  CODE        ENDS
92              END      START
```

实验四 最小拍控制系统

接线参考

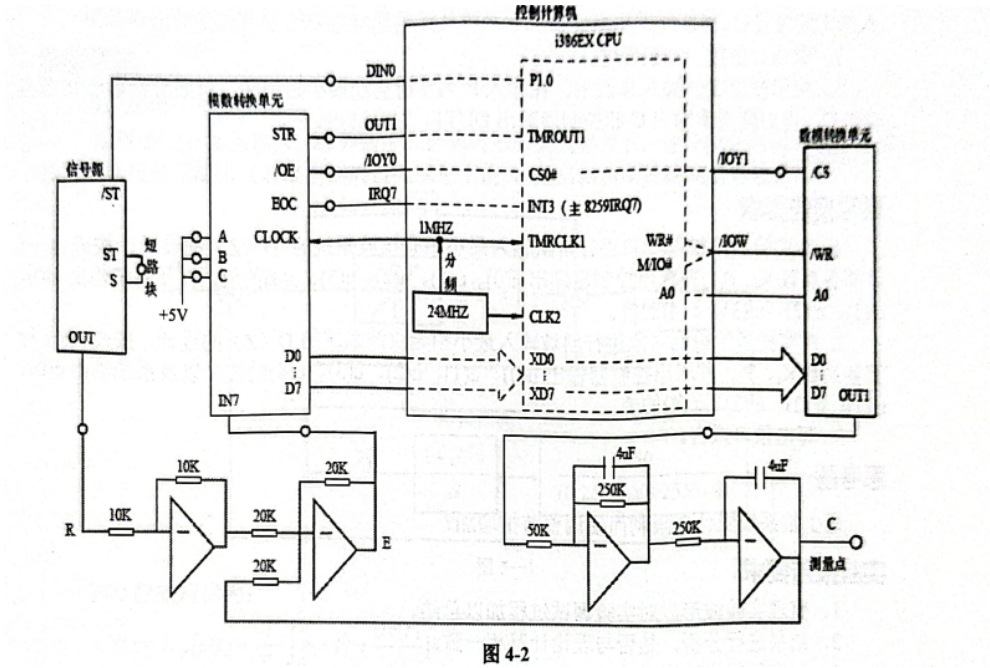
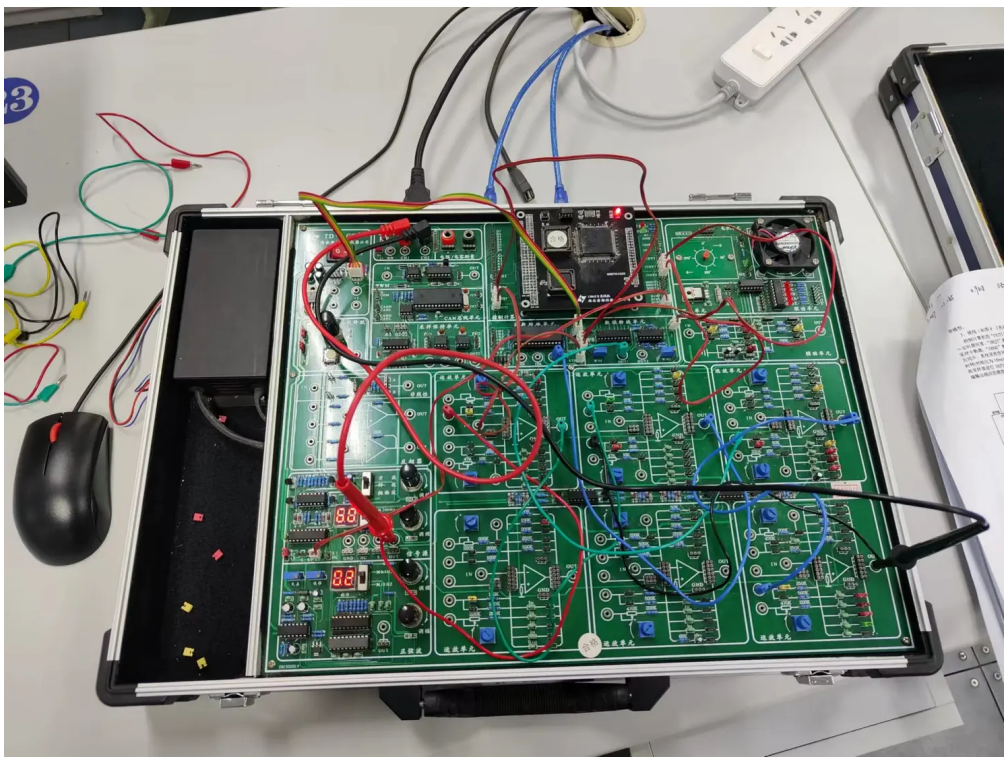


图 4-2



有纹波

Plain Text

```
;最小拍算法（有纹波）

AD_ADDR   EQU    0600H
DA1_ADDR  EQU    0640H
DIN_ADDR  EQU   0F860H

STACK1          SEGMENT       STACK
                DW            256 DUP(?)
STACK1          ENDS

DATA            SEGMENT
SAVEBUFF        DB       256 DUP(00H)     ;存储缓冲
TK              DB    64H                 ;采样周期
K0              DB     00H,54H,35H             ;系数K0的值为+0.5435
K1              DB     01H,20H,00H             ;系数K1的值为-0.2000
K2              DB     00H,00H,00H             ;系数K2的值为0
K3              DB     00H,00H,00H             ;系数K3的值为0
P1              DB     00H,71H,70H             ;系数P1的值为+0.7180
P2              DB     00H,00H,00H             ;系数P2的值为0
P3              DB     00H,00H,00H             ;系数P3的值为0
SAVEMARK        DB    00H                    ;存储缓冲标志
SAVEADDR        DB    0FFH                  ;存储缓冲相加标志
CONTROLMARK     DB    00H                    ;存储缓冲控制标志
EKB             DB    00H                    ;偏差（字节）
K0_16           DB    00H,00H,00H            ;系数K0转换二进制小数的变量
K1_16           DB    00H,00H,00H            ;系数K1转换二进制小数的变量
K2_16           DB    00H,00H,00H            ;系数K2转换二进制小数的变量
K3_16           DB    00H,00H,00H            ;系数K3转换二进制小数的变量
P1_16           DB    00H,00H,00H            ;系数P1转换二进制小数的变量
P2_16           DB    00H,00H,00H            ;系数P2转换二进制小数的变量
P3_16           DB    00H,00H,00H            ;系数P3转换二进制小数的变量
EK              DW    0000H                 ;偏差（字）
EK_1            DW    0000H                 ;上一次偏差
EK_2            DW    0000H                 ;上上一次偏差
EK_3            DW    0000H                 ;上上上一次偏差
UK              DW    0000H                 ;当前的控制量
UK_1               DW    0000H                  ;上一次的控制量
UK_2            DW    0000H                 ;上上一次控制量
UK_3            DW    0000H                 ;上上上一次的控制量
OUTPUT          DW    0000H                 ;控制量输出的变量
ALLK_ALLP       DW    0000H                 ;K1*EK_1+K2*EK_2+K3*EK_3-P1*
UK_1-P2*UK_2-P3*UK_3的值

```

```
43 DATA      ENDS
44
45 CODE      SEGMENT
46           ASSUME        CS:CODE,DS:DATA
47
48 START:  MOV       AX,DATA                ;填写数据段
49         MOV       DS,AX
50
51           PUSH    DS
52           XOR        AX,AX                ;填写7号中断入口地址
53           MOV        DS,AX
54         MOV       AX,OFFSET IRQ7
55         MOV       SI,003CH
56         MOV       [SI],AX
57         MOV       AX,CS
58         MOV       SI,003EH
59         MOV       [SI],AX
60         CLI
61           POP        DS
62
63         CALL     SYSINTI                ;系统初始化
64         MOV          DX,0F043H             ;初始化定时器1 定时10ms

65         MOV     AL,076H
66         OUT     DX,AL
67         MOV          DX,0F041H
68         MOV     AL,10H
69         OUT     DX,AL
70         MOV          DX,0F041H
71         MOV     AL,27H
72         OUT     DX,AL
73         IN         AL,21H              ;开7号中断
74         AND        AL,7FH
75         OUT        21H,AL
76         MOV        SI,OFFSET P3+2        ;系数转换成二进制小数

77         MOV        BH,07H
78         MOV        DI,OFFSET P3_16+2
79         CALL       CHANGE
80         CALL       CLEAR_E             ;变量清零
81
82         MOV          DX,DA1_ADDR
83         MOV     AL,80H                 ;D/A0832置零
84         OUT     DX,AL
85         MOV        BL,01H
86         MOV        SAVEMARK,00H
```

```
 87          MOV       SAVEADDR,0FFH
 88  AGAIN:  STI
 89          HLT
 90          JMP       AGAIN
 91
 92  IRQ7:   MOV          DX,DIN_ADDR            ;判同步信号到否
 93          IN          AL,DX
 94          AND          AL,01H
 95          JZ          ZZB
 96          MOV          DX,DA1_ADDR
 97          MOV     AL,80H                ;D/A0832置零
 98          OUT     DX,AL
 99
100          MOV       CONTROLMARK,AL
101          MOV       SAVEMARK,AL
102          CALL      CLEAR_E
103          MOV       BL,01H
104          JMP       FINISH
105
106 ZZB:       MOV       AL,01H
107          MOV       CONTROLMARK,AL
108          DEC       BL
109          JNZ       FINISH
110
111          CALL      CY                      ;调用采样子程序
112          MOV       EK,DX
113          MOV       AX,DX                 ;计算K0*EK
114          MOV       DI,OFFSET K0_16
115          INC     DI
116          MOV       DX,[DI]
117          CALL      ML
118          MOV       AX,ALLK_ALLP          ;与变量ALLK_ALLP相加，得到控制量
119          ADD       DX,AX
120          CALL      OUT_PUT
121          MOV       DX,OUTPUT
122          MOV       UK,DX                 ;EK,EK_1,EK_2,EK_3,UK,UK_1,UK_2,U
    K_3变量递推
123          MOV       SI,OFFSET UK_2
124          MOV       DI,OFFSET UK_3
125          MOV       BL,07H
126 L1:      MOV       AX,[SI]
127          MOV       [DI],AX
128          DEC       SI
129          DEC       SI
130          DEC       DI
131            DEC        DI
```

```
132         DEC       BL
133         JNZ       L1
134         MOV       AX,0000H              ;计算K1*EK_1+K2*EK_2+K3*EK_3-P1
    *UK_1-P2*UK_2-P3*UK_3
135         MOV       ALLK_ALLP,AX
136         MOV       SI,OFFSET K1_16
137         MOV       DI,OFFSET EK_1
138         MOV       CX,0003H
139         CALL      L2
140         MOV       SI,OFFSET P1_16
141         MOV       DI,OFFSET UK_1
142         MOV       CX,0103H
143         CALL      L2
144         MOV       BL,TK                 ;采样周期变量还原

145 FINISH: MOV       AL,20H                ;中断返回
146         OUT       20H,AL
147         IRET
148
149 L2:     INC     SI
150         MOV       DX,[SI]
151         MOV       AX,[DI]
152         CALL     ML
153         DEC       SI
154         TEST      BYTE PTR[SI],01H
155         JNZ       QB
156 LL:     INC       SI
157         INC       SI
158         INC       SI
159         INC       DI
160         INC       DI
161
162         MOV       AX,ALLK_ALLP
163         CMP       CH,00H
164         JNZ       L3
165         ADD       DX,AX
166 L4:     MOV       ALLK_ALLP,DX
167         DEC       CL
168         JNZ       L2
169         RET
170
171 L3:     XCHG      DX,AX
172         SUB       DX,AX
173         JMP       L4
174 QB:     NEG       DX
175         JMP       LL
```

```
176
177 ;乘法子程序,
178 ;适用范围： 被乘数为符号数，乘数为无符号数，16位乘法
179 ;参数：      被乘数AX，乘数DX
180 ;返回值：     结果的高16位在DX中，低16位在AX中
181 ML:         CMP     DX,7FFFH
182          JA         ML1
183          IMUL     DX
184          RET
185 ML1:       PUSH     BX
186          PUSH     CX
187          PUSH      AX
188          SUB       DX,7FFFH
189          IMUL     DX
190          MOV       BX,DX
191          MOV       CX,AX
192          POP        AX
193          MOV       DX,7FFFH
194          IMUL      DX
195          ADD        AX,CX
196          ADC        DX,BX
197          POP        CX
198          POP        BX
199          RET
200
201 ;采样子程序
202 ;功能：          从A/D读取采样值，并进行变换，采样值大于80H，表示0～－5V，采样值小于
    80H,表示0～＋5V
203 ;              并计算偏差EK，分别将其扩大32倍，为了保证计算精度
204 ;返回值：     返回值存在相应的变量中
205 CY:      MOV           DX,AD_ADDR
206          IN       AL,DX
207          SUB        AL,80H
208          MOV       EKB,AL
209          CALL      SAVEEK
210          MOV       DX,0000H
211          MOV       DH,EKB
212          CLC
213          MOV       CL,03H
214          SAR       DX,CL
215          RET
216
217 SAVEEK: MOV       AH,AL
218          MOV       DX,OFFSET SAVEBUFF
219          MOV       AL,CONTROLMARK
220          CMP       AL,00H
```

```
221         JZ        L5
222         MOV       AL,SAVEMARK
223         CMP       AL,01H
224         JZ        L5
225         MOV       AL,SAVEADDR
226         CMP       AL,0FEH
227         JZ        L6
228         INC       AL
229         MOV       DL,AL
230         MOV       SAVEADDR,AL
231         PUSH      DI
232         MOV       DI,DX
233         MOV       [DI],AH
234         POP      DI
235 L5:     MOV       AL,AH
236         RET
237 L6:     INC       SAVEMARK
238         JMP       L5
239
240 ;控制量输出子程序
241 ;功能：         控制量判溢出，放大8倍后输出
242 OUT_PUT:MOV       OUTPUT,DX
243         TEST      DH,80H                ;判溢出
244         JZ        L7
245         MOV       BX,0F000H
246         SUB       DX,BX
247         JG        L8
248         MOV       DX,0F000H
249         JMP       L9
250 L7:     MOV       BX,0FFFH
251         SUB       DX,BX
252         JG        L10
253 L8:     MOV       DX,OUTPUT
254         JMP       L11
255 L10:    MOV       DX,0FFFH
256
257 L9:     MOV       OUTPUT,DX
258 L11:    MOV       CL,03H
259         MOV       AL,CONTROLMARK
260         CMP       AL,00H
261         JZ        L12
262         SHL       DX,CL
263         MOV       AL,DH                 ;控制量输出
264         MOV     DX,DA1_ADDR
265         ADD     AL,80H
266         OUT     DX,AL
```

```
267 L12:    RET
268
269 ;系数转换子程序
270 ;功能：          将10进制表示的数转换成2进制小数，并保留其符号
271 ;返回值;     转换结果保存在相应的变量中
272 CHANGE: DEC       SI
273         MOV       CX,[SI]
274         INC       SI
275         AND       AL,AL
276         MOV       DX,0000H
277         MOV       BL,10H
278 GO:     MOV       AL,[SI]
279         ADD       AL,AL
280         DAA
281         MOV       [SI],AL
282         DEC       SI
283         MOV       AL,[SI]
284         ADC      AL,AL
285         DAA
286         RCL       DX,0001H
287         MOV       [SI],AL
288         INC       SI
289         DEC       BL
290         JNZ       GO
291         DEC       SI
292         MOV       [SI],CX
293         DEC       SI
294         MOV       AL,[SI]
295         DEC       DI
296         MOV       [DI],DX
297         DEC       SI
298         DEC       DI
299         MOV       [DI],AL
300         DEC       DI
301         DEC       BH
302         JNZ       CHANGE
303         RET
304
305 ;变量清零子程序
306 CLEAR_E:MOV       AX,0000H
307         MOV       ALLK_ALLP,AX
308         MOV       SI,OFFSET EK
309         MOV       BL,08H
310 GOON:   MOV       [SI],AX
311         INC       SI
312         INC       SI
```

```
313         DEC       BL
314         JNZ       GOON
315         RET
316
317 ;延时子程序
318 ;功能：         延时一段时间
319 ;无返回值
320 DELAY:  PUSH      CX
321         MOV       CX,1000H
322 DEL1:   PUSH      AX
323         POP       AX
324         LOOP      DEL1
325         POP       CX
326         RET
327
328 ;系统初始化子程序
329 ;功能：         初始化IO,和CS0片选，管脚配置，中断
330 sysinti:MOV       AX,8000h
331            OUT       23h,AL
332               XCHG      AL,AH
333            OUT       22H,AL
334         OUT       22H,AX
335
336           MOV       DX,0F822H
337         MOV        AL,70H
338         OUT        DX,AL
339         MOV        DX,0F824H
340         MOV        AL,0B2H
341         OUT        DX,AL
342         MOV        DX,0F832H
343         MOV        AL,0AH
344         OUT        DX,AL
345         MOV        DX,0F834H
346         MOV        AL,1DH
347         OUT        DX,AL
348
349           MOV       AL,11H
350         OUT        20H,AL
351         MOV        AL,08H
352         OUT        21H,AL
353         MOV        AL,04H
354         OUT        21H,AL
355         MOV        AL,01H
356         OUT        21H,AL
357         MOV        AL,0EFH
358         OUT        21H,AL
```

```
359
360          MOV         AL,11H
361          OUT         0A0H,AL
362          MOV         AL,30H
363          OUT         0A1H,AL
364          MOV         AL,02H
365          OUT         0A1H,AL
366          MOV         AL,01H
367          OUT         0A1H,AL
368          MOV         AL,0FFH
369          OUT         0A1H,AL
370          RET
371
372  CODE    ENDS
373          END         START
374
```

无纹波

Plain Text

```
;最小拍算法（无纹波）

AD_ADDR   EQU    0600H
DA1_ADDR  EQU    0640H
DIN_ADDR  EQU   0F860H

STACK1          SEGMENT        STACK
                DW             256 DUP(?)
STACK1          ENDS

DATA            SEGMENT
SAVEBUFF        DB        256 DUP(00H)     ;存储缓冲
TK              DB     64H                 ;采样周期
K0              DB      00H,76H,50H             ;系数K0的值为+0.7650
K1              DB      01H,73H,20H             ;系数K1的值为-0.7320
K2              DB      00H,16H,51H             ;系数K2的值为+0.1651
K3              DB      00H,00H,00H             ;系数K3的值为0
P1              DB      01H,40H,80H             ;系数P1的值为-0.4080
P2              DB      01H,59H,20H             ;系数P2的值为-0.5920
P3              DB      00H,00H,00H             ;系数P3的值为0
SAVEMARK        DB     00H                     ;存储缓冲标志
SAVEADDR        DB     0FFH                   ;存储缓冲相加标志
CONTROLMARK     DB     00H                     ;存储缓冲控制标志
EKB             DB     00H                     ;偏差（字节）
K0_16           DB     00H,00H,00H             ;系数K0转换二进制小数的变量
K1_16           DB     00H,00H,00H             ;系数K1转换二进制小数的变量
K2_16           DB     00H,00H,00H             ;系数K2转换二进制小数的变量
K3_16           DB     00H,00H,00H             ;系数K3转换二进制小数的变量
P1_16           DB     00H,00H,00H             ;系数P1转换二进制小数的变量
P2_16           DB     00H,00H,00H             ;系数P2转换二进制小数的变量
P3_16           DB     00H,00H,00H             ;系数P3转换二进制小数的变量
EK              DW     0000H                  ;偏差（字）
EK_1            DW     0000H                  ;上一次偏差
EK_2            DW     0000H                  ;上上一次偏差
EK_3            DW     0000H                  ;上上上一次偏差
UK              DW     0000H                  ;当前的控制量
UK_1               DW    0000H                   ;上一次的控制量
UK_2            DW     0000H                  ;上上一次控制量
UK_3            DW     0000H                  ;上上上一次的控制量
OUTPUT          DW     0000H                  ;控制量输出的变量
ALLK_ALLP       DW     0000H                  ;K1*EK_1+K2*EK_2+K3*EK_3-P1*
UK_1-P2*UK_2-P3*UK_3的值

```

```
43 DATA      ENDS
44
45 CODE      SEGMENT
46           ASSUME        CS:CODE,DS:DATA
47
48 START:  MOV       AX,DATA                ;填写数据段
49         MOV       DS,AX
50
51           PUSH    DS
52           XOR        AX,AX                ;填写7号中断入口地址
53           MOV        DS,AX
54         MOV       AX,OFFSET IRQ7
55         MOV       SI,003CH
56         MOV       [SI],AX
57         MOV       AX,CS
58         MOV       SI,003EH
59         MOV       [SI],AX
60         CLI
61           POP        DS
62
63         CALL     SYSINTI                ;系统初始化
64         MOV          DX,0F043H             ;初始化定时器1 定时10ms

65         MOV     AL,076H
66         OUT     DX,AL
67         MOV          DX,0F041H
68         MOV     AL,10H
69         OUT     DX,AL
70         MOV          DX,0F041H
71         MOV     AL,27H
72         OUT     DX,AL
73         IN         AL,21H              ;开7号中断
74         AND        AL,7FH
75         OUT        21H,AL
76         MOV        SI,OFFSET P3+2        ;系数转换成二进制小数

77         MOV        BH,07H
78         MOV        DI,OFFSET P3_16+2
79         CALL       CHANGE
80         CALL       CLEAR_E             ;变量清零
81
82         MOV          DX,DA1_ADDR
83         MOV     AL,80H                 ;D/A0832置零
84         OUT     DX,AL
85         MOV        BL,01H
86         MOV        SAVEMARK,00H
```

```
 87          MOV       SAVEADDR,0FFH
 88  AGAIN:  STI
 89          HLT
 90          JMP       AGAIN
 91
 92  IRQ7:   MOV          DX,DIN_ADDR            ;判同步信号到否
 93          IN          AL,DX
 94          AND          AL,01H
 95          JZ          ZZB
 96          MOV          DX,DA1_ADDR
 97          MOV     AL,80H                ;D/A0832置零
 98          OUT     DX,AL
 99
100          MOV       CONTROLMARK,AL
101          MOV       SAVEMARK,AL
102          CALL      CLEAR_E
103          MOV       BL,01H
104          JMP       FINISH
105
106 ZZB:       MOV       AL,01H
107          MOV       CONTROLMARK,AL
108          DEC       BL
109          JNZ       FINISH
110
111          CALL      CY                      ;调用采样子程序
112          MOV       EK,DX
113          MOV       AX,DX                 ;计算K0*EK
114          MOV       DI,OFFSET K0_16
115          INC     DI
116          MOV       DX,[DI]
117          CALL      ML
118          MOV       AX,ALLK_ALLP          ;与变量ALLK_ALLP相加，得到控制量
119          ADD       DX,AX
120          CALL      OUT_PUT
121          MOV       DX,OUTPUT
122          MOV       UK,DX                 ;EK,EK_1,EK_2,EK_3,UK,UK_1,UK_2,U
    K_3变量递推
123          MOV       SI,OFFSET UK_2
124          MOV       DI,OFFSET UK_3
125          MOV       BL,07H
126 L1:      MOV       AX,[SI]
127          MOV       [DI],AX
128          DEC       SI
129          DEC       SI
130          DEC       DI
131            DEC        DI
```

```
132         DEC       BL
133         JNZ       L1
134         MOV       AX,0000H              ;计算K1*EK_1+K2*EK_2+K3*EK_3-P1
    *UK_1-P2*UK_2-P3*UK_3
135         MOV       ALLK_ALLP,AX
136         MOV       SI,OFFSET K1_16
137         MOV       DI,OFFSET EK_1
138         MOV       CX,0003H
139         CALL      L2
140         MOV       SI,OFFSET P1_16
141         MOV       DI,OFFSET UK_1
142         MOV       CX,0103H
143         CALL      L2
144         MOV       BL,TK                 ;采样周期变量还原

145 FINISH: MOV       AL,20H                ;中断返回
146         OUT       20H,AL
147         IRET
148
149 L2:     INC     SI
150         MOV       DX,[SI]
151         MOV       AX,[DI]
152         CALL     ML
153         DEC       SI
154         TEST      BYTE PTR[SI],01H
155         JNZ       QB
156 LL:     INC       SI
157         INC       SI
158         INC       SI
159         INC       DI
160         INC       DI
161
162         MOV       AX,ALLK_ALLP
163         CMP       CH,00H
164         JNZ       L3
165         ADD       DX,AX
166 L4:     MOV       ALLK_ALLP,DX
167         DEC       CL
168         JNZ       L2
169         RET
170
171 L3:     XCHG      DX,AX
172         SUB       DX,AX
173         JMP       L4
174 QB:     NEG       DX
175         JMP       LL
```

```
176
177 ;乘法子程序,
178 ;适用范围： 被乘数为符号数，乘数为无符号数，16位乘法
179 ;参数：      被乘数AX，乘数DX
180 ;返回值：     结果的高16位在DX中，低16位在AX中
181 ML:         CMP     DX,7FFFH
182          JA         ML1
183          IMUL     DX
184          RET
185 ML1:       PUSH     BX
186          PUSH     CX
187          PUSH      AX
188          SUB       DX,7FFFH
189          IMUL     DX
190          MOV       BX,DX
191          MOV       CX,AX
192          POP        AX
193          MOV       DX,7FFFH
194          IMUL      DX
195          ADD        AX,CX
196          ADC        DX,BX
197          POP        CX
198          POP        BX
199          RET
200
201 ;采样子程序
202 ;功能：          从A/D读取采样值，并进行变换，采样值大于80H，表示0～－5V，采样值小于
    80H,表示0～＋5V
203 ;              并计算偏差EK，分别将其扩大32倍，为了保证计算精度
204 ;返回值：     返回值存在相应的变量中
205 CY:      MOV           DX,AD_ADDR
206          IN       AL,DX
207          SUB        AL,80H
208          MOV       EKB,AL
209          CALL      SAVEEK
210          MOV       DX,0000H
211          MOV       DH,EKB
212          CLC
213          MOV       CL,03H
214          SAR       DX,CL
215          RET
216
217 SAVEEK: MOV       AH,AL
218          MOV       DX,OFFSET SAVEBUFF
219          MOV       AL,CONTROLMARK
220          CMP       AL,00H
```

```
221         JZ        L5
222         MOV       AL,SAVEMARK
223         CMP       AL,01H
224         JZ        L5
225         MOV       AL,SAVEADDR
226         CMP       AL,0FEH
227         JZ        L6
228         INC       AL
229         MOV       DL,AL
230         MOV       SAVEADDR,AL
231         PUSH      DI
232         MOV       DI,DX
233         MOV       [DI],AH
234         POP      DI
235 L5:     MOV       AL,AH
236         RET
237 L6:     INC       SAVEMARK
238         JMP       L5
239
240 ;控制量输出子程序
241 ;功能：         控制量判溢出，放大8倍后输出
242 OUT_PUT:MOV       OUTPUT,DX
243         TEST      DH,80H                ;判溢出
244         JZ        L7
245         MOV       BX,0F000H
246         SUB       DX,BX
247         JG        L8
248         MOV       DX,0F000H
249         JMP       L9
250 L7:     MOV       BX,0FFFH
251         SUB       DX,BX
252         JG        L10
253 L8:     MOV       DX,OUTPUT
254         JMP       L11
255 L10:    MOV       DX,0FFFH
256
257 L9:     MOV       OUTPUT,DX
258 L11:    MOV       CL,03H
259         MOV       AL,CONTROLMARK
260         CMP       AL,00H
261         JZ        L12
262         SHL       DX,CL
263         MOV       AL,DH                 ;控制量输出
264         MOV     DX,DA1_ADDR
265         ADD     AL,80H
266         OUT     DX,AL
```

```
267 L12:    RET
268
269 ;系数转换子程序
270 ;功能：          将10进制表示的数转换成2进制小数，并保留其符号
271 ;返回值;     转换结果保存在相应的变量中
272 CHANGE: DEC       SI
273         MOV       CX,[SI]
274         INC       SI
275         AND       AL,AL
276         MOV       DX,0000H
277         MOV       BL,10H
278 GO:     MOV       AL,[SI]
279         ADD       AL,AL
280         DAA
281         MOV       [SI],AL
282         DEC       SI
283         MOV       AL,[SI]
284         ADC      AL,AL
285         DAA
286         RCL       DX,0001H
287         MOV       [SI],AL
288         INC       SI
289         DEC       BL
290         JNZ       GO
291         DEC       SI
292         MOV       [SI],CX
293         DEC       SI
294         MOV       AL,[SI]
295         DEC       DI
296         MOV       [DI],DX
297         DEC       SI
298         DEC       DI
299         MOV       [DI],AL
300         DEC       DI
301         DEC       BH
302         JNZ       CHANGE
303         RET
304
305 ;变量清零子程序
306 CLEAR_E:MOV       AX,0000H
307         MOV       ALLK_ALLP,AX
308         MOV       SI,OFFSET EK
309         MOV       BL,08H
310 GOON:   MOV       [SI],AX
311         INC       SI
312         INC       SI
```

```
313         DEC       BL
314         JNZ       GOON
315         RET
316
317 ;延时子程序
318 ;功能：         延时一段时间
319 ;无返回值
320 DELAY:  PUSH      CX
321         MOV       CX,1000H
322 DEL1:   PUSH      AX
323         POP       AX
324         LOOP      DEL1
325         POP       CX
326         RET
327
328 ;系统初始化子程序
329 ;功能：         初始化IO,和CS0片选，管脚配置，中断
330 sysinti:MOV       AX,8000h
331            OUT       23h,AL
332               XCHG      AL,AH
333            OUT       22H,AL
334         OUT       22H,AX
335
336           MOV       DX,0F822H
337         MOV        AL,70H
338         OUT        DX,AL
339         MOV        DX,0F824H
340         MOV        AL,0B2H
341         OUT        DX,AL
342         MOV        DX,0F832H
343         MOV        AL,0AH
344         OUT        DX,AL
345         MOV        DX,0F834H
346         MOV        AL,1DH
347         OUT        DX,AL
348
349           MOV       AL,11H
350         OUT        20H,AL
351         MOV        AL,08H
352         OUT        21H,AL
353         MOV        AL,04H
354         OUT        21H,AL
355         MOV        AL,01H
356         OUT        21H,AL
357         MOV        AL,0EFH
358         OUT        21H,AL
```

```
359
360         MOV         AL,11H
361         OUT         0A0H,AL
362         MOV         AL,30H
363         OUT         0A1H,AL
364         MOV         AL,02H
365         OUT         0A1H,AL
366         MOV         AL,01H
367         OUT         0A1H,AL
368         MOV         AL,0FFH
369         OUT         0A1H,AL
370         RET
371
372 CODE    ENDS
373         END        START
374
```